



BOARD OF INTERMEDIATE & SECONDARY EDUCATION, LARKANO

Established to serve affiliated schools and colleges across the Larkano region

Higher Secondary School Certificate
Examination Syllabus

BIOLOGY

Grades XI – XII

This syllabus will be examined in both Annual and Re-sit Examination sessions beginning with the Annual Examinations 2023.

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Preface

The Board of Intermediate & Secondary Education, Larkano (BISE Larkano) is a public examination body responsible for the conduct of Secondary School Certificate (SSC) and Higher Secondary School Certificate (HSSC) examinations for its affiliated schools and colleges. The Board is committed to developing examination syllabi that promote conceptual understanding and scientific reasoning, and that remain aligned with the national curriculum and with contemporary international standards in science education.

This document sets out the syllabus for Biology at the HSSC level (Grades XI and XII). It has been developed in recognition of biology's role as a foundation for further study and employment in medicine, pharmacy, agriculture, biotechnology, environmental science and a wide range of other life-science fields.

The syllabus was developed following consultation with subject specialists, teachers of affiliated institutions, and reviewers with expertise in biology education, and reflects a deliberate effort to move examination preparation away from rote memorisation of facts and toward careful observation, experimentation and biological reasoning.

We are confident that this syllabus will support teachers and students of affiliated institutions in building a rigorous and practical understanding of biology, and we wish every success to those who study and teach it.

Controller of Examinations

Board of Intermediate & Secondary Education, Larkano

Understanding of BISE Larkano Syllabi

1. The BISE Larkano syllabi guide students, teachers, parents and other stakeholders regarding the topics that will be taught and examined in each grade (XI and XII). In each syllabus document, the content progresses from simple to complex, facilitating gradual, conceptual learning.
2. The topics of each syllabus are divided into sub-topics and Student Learning Outcomes (SLOs). The sub-topics and SLOs define the depth and breadth at which each topic will be taught, learnt and examined.
3. Each SLO begins with an achievable, assessable command word such as ‘describe’, ‘apply’, ‘analyse’ or ‘evaluate’. Examination questions are framed using the same command words, or their equivalents, to elicit evidence of these competencies in students’ responses.
4. The SLOs are classified under three cognitive levels: Knowledge (K), Understanding (U), and Application and other higher-order skills (A), to support effective planning by teachers and the balanced construction of examination papers.
5. By focusing on the achievement of SLOs rather than on the reproduction of memorised facts, this syllabus aims to encourage students to reason scientifically, to justify their conclusions with evidence and calculation, and to communicate their thinking clearly — skills that extend well beyond a single assignment or examination.
6. While the syllabus recommends locally available textbooks and resources for achieving these outcomes, teachers and students are encouraged to draw on multiple teaching and learning resources, and to acknowledge any resource that materially shapes a solution submitted for assessment.

Subject Rationale of Biology

What will you learn in Biology?

Biology is the study of living organisms — their structure, function, diversity and the processes by which they grow, reproduce and interact with their environment. This syllabus teaches students to explain biological phenomena in terms of underlying structures and processes, and to support their explanations with careful observation and experimental evidence rather than with memorised statements.

Many students who can recall a definition have not yet learned to apply it — to interpret a set of experimental results, to work through a genetics problem, or to explain how a change in one part of a living system affects another. This syllabus addresses that gap directly. Students will study the molecular and cellular basis of life, the physiology of plants and animals, the mechanisms of heredity and molecular biology, and the principles of evolution and ecology.

Throughout, the syllabus emphasises the application of biological principles to unfamiliar situations, and the ability to reason from evidence — in the laboratory, in the field and in the examination room — rather than the recall of isolated facts.

Where will it take you?

The foundation provided by this syllabus prepares students for further study in Biology, Medicine, Pharmacy, Biotechnology and related disciplines, and for a wide range of careers, including:

- Doctor / Surgeon
- Pharmacist
- Biotechnologist
- Microbiologist
- Genetic Counsellor
- Environmental or Wildlife Scientist
- Agricultural Scientist
- Veterinarian

How to approach the syllabus?

The concept map on the following page gives an overview of the entire syllabus. The topics and Student Learning Outcomes guide what has to be achieved, and the Scheme of Assessment explains how each topic will be weighted and examined.

Concept Map

The concept map below organises the syllabus into four connected strands, all resting on a foundation of Observation, Experimentation and Scientific Reasoning — the investigative skills that make every other strand of biology possible.

Molecules and Cells

- Biological Molecules
- Enzymes
- Cell Structure and Membrane Transport
- The Cell Cycle, Mitosis and Meiosis

Energy and Physiology

- Photosynthesis and Respiration
- Homeostasis and Transport
- Support and Movement
- Nervous and Hormonal Coordination

Continuity of Life

- Reproduction
- Growth and Development
- Mendelian Genetics
- Molecular Biology and Biotechnology

Diversity and Environment

- Classification and Biodiversity
- Evolution by Natural Selection
- Ecosystems and Energy Flow
- Human Impact and Conservation

Observation, Experimentation and Scientific Reasoning underpin all four strands: every dissection, every data set and every genetics problem in this syllabus is ultimately a tool for reasoning carefully from evidence to a defensible conclusion.

Student Learning Outcomes of Biology

Part I (Grade XI)

1. Introduction to Biology and the Diversity of Life

Topics and Sub-topics	Student Learning Outcomes	K	U	A
1.1 Nature and Scope of Biology	describe biology as the scientific study of living organisms and outline its major branches;	✓		
	distinguish between the characteristics that define a living organism and non-living matter;		✓	
	describe the steps of the scientific method with reference to a biological investigation;	✓		
1.2 Classification and Biodiversity	describe the hierarchical system of biological classification from kingdom to species;	✓		
	outline the principal features used to distinguish the five kingdoms of living organisms;		✓	
	explain the importance of biodiversity and the need for its conservation;		✓	

2. Biological Molecules and Enzymes

Topics and Sub-topics	Student Learning Outcomes	K	U	A
2.1 Biological Molecules	describe the general structure of carbohydrates, lipids, proteins and nucleic acids;	✓		
	relate the structure of a given biomolecule to its biological function;		✓	
	test a sample for the presence of a named biomolecule using a standard laboratory test;			✓
2.2 Enzymes	describe the general properties of enzymes as biological catalysts;	✓		
	explain the effect of temperature, pH and substrate concentration on the rate of an enzyme-catalysed reaction;			✓
	interpret data from an experiment investigating enzyme activity;			✓

3. Cell Structure and Function

Topics and Sub-topics	Student Learning Outcomes	K	U	A
3.1 The Cell as the Unit of Life	describe the structure of a generalised prokaryotic cell and a generalised eukaryotic cell;	✓		
	distinguish between the structure of a typical plant cell and a typical animal cell;		✓	
	relate the structure of a named cell organelle to its function;		✓	
3.2 Cell Membrane and Transport	describe the fluid mosaic model of the cell membrane;	✓		
	distinguish between diffusion, osmosis and active transport;		✓	
	predict the effect of a hypotonic, hypertonic or isotonic solution on a plant or animal cell;			✓

4. Cell Division and Reproduction at the Cellular Level

Topics and Sub-topics	Student Learning Outcomes	K	U	A
4.1 The Cell Cycle	describe the stages of the cell cycle;	✓		
	explain the significance of checkpoints in the regulation of the cell cycle;		✓	
4.2 Mitosis and Meiosis	describe the sequence of events in mitosis;	✓		
	describe the sequence of events in meiosis;	✓		
	distinguish between the outcomes and the biological significance of mitosis and meiosis;		✓	
	interpret a diagram or photomicrograph to identify a given stage of cell division;			✓

5. Bioenergetics

Topics and Sub-topics	Student Learning Outcomes	K	U	A
5.1 Photosynthesis	describe the overall equation for photosynthesis and identify its raw materials and products;	✓		
	outline the light-dependent and light-independent stages of photosynthesis;		✓	
	interpret experimental data on the effect of light intensity, carbon dioxide concentration or temperature on the rate of photosynthesis;			✓
5.2 Cellular Respiration	describe the overall equation for aerobic cellular respiration;	✓		
	distinguish between aerobic and anaerobic respiration;		✓	
	interpret experimental data on the rate of respiration in a living organism;			✓

Part II (Grade XII)

6. Homeostasis and Transport

Topics and Sub-topics	Student Learning Outcomes	K	U	A
6.1 Homeostasis	define homeostasis and explain its importance to a living organism;	✓		
	describe the role of a named organ system in maintaining a stable internal environment;		✓	
6.2 Transport in Plants and Animals	describe the pathway of water and mineral transport in a vascular plant;	✓		
	describe the structure and function of the mammalian circulatory system;	✓		
	relate the structure of a blood vessel to its function;		✓	

7. Support, Movement and Coordination

Topics and Sub-topics	Student Learning Outcomes	K	U	A
7.1 Support and Movement	describe the structure and function of a named skeletal system;	✓		
	explain the role of muscles and joints in producing movement;		✓	
7.2 Nervous and Hormonal Coordination	describe the structure and function of a neuron;	✓		
	describe the transmission of a nerve impulse across a synapse;		✓	
	distinguish between nervous and hormonal coordination in terms of the speed and duration of response;		✓	
	explain how a named hormone regulates a specific physiological process;			✓

8. Reproduction and Development

Topics and Sub-topics	Student Learning Outcomes	K	U	A
8.1 Reproduction	distinguish between asexual and sexual reproduction;		✓	
	describe the structure and function of the human male and female reproductive systems;	✓		
8.2 Growth and Development	describe the stages of growth and development in a named organism;	✓		
	interpret a growth curve to identify the phases of growth it represents;			✓

9. Genetics and Molecular Biology

Topics and Sub-topics	Student Learning Outcomes	K	U	A
9.1 Principles of Inheritance	describe Mendel's laws of inheritance;	✓		
	solve a genetics problem involving monohybrid or dihybrid inheritance using a Punnett square;			✓
	explain a pattern of inheritance that deviates from simple Mendelian ratios, e.g. codominance or sex linkage;		✓	
9.2 Molecular Basis of Inheritance and Biotechnology	describe the structure of DNA and explain its role in storing genetic information;	✓		
	outline the process of protein synthesis, i.e. transcription and translation;		✓	

Topics and Sub-topics	Student Learning Outcomes	K	U	A
	describe an application of biotechnology, e.g. genetic engineering or selective breeding, and evaluate its benefits and risks;			✓

10. Evolution and Ecology

Topics and Sub-topics	Student Learning Outcomes	K	U	A
10.1 Evolution	describe the theory of evolution by natural selection;	✓		
	interpret evidence for evolution from a named source, e.g. fossils, comparative anatomy or molecular data;			✓
10.2 Ecology and Environment	describe the flow of energy through a food chain and a food web;	✓		
	describe a named nutrient cycle, e.g. the carbon cycle or the nitrogen cycle;	✓		
	explain the impact of human activity on a named ecosystem and evaluate a strategy for its conservation;			✓

Scheme of Assessment

Each grade of Biology is examined through a Theory paper and a Practical (laboratory) examination, whose marks combine to the 100 marks recorded for the subject on the BISE Larkano Marks Certificate. The Theory paper carries 85 marks and the Practical examination carries 15 marks, matching the maximum marks shown against ‘Biology (TH)’ and ‘Biology (PR)’ for each year on the certificate.

Grade XI

Table 1: Number of Student Learning Outcomes by Cognitive Level

Topic No.	Topic	No. of Sub-topics	K	U	A	Total
1	Introduction to Biology and the Diversity of Life	2	3	3	0	6
2	Biological Molecules and Enzymes	2	2	1	3	6
3	Cell Structure and Function	2	2	3	1	6
4	Cell Division and Reproduction at the Cellular Level	2	3	2	1	6
5	Bioenergetics	2	2	2	2	6
	Total	10	12	11	7	30
	Percentage		40%	37%	23%	100%

Table 2: Theory Exam Specification — Grade XI — Biology (TH)-I — 85 Marks

Topic No.	Topic	MCQs (Marks)	CRQs (Marks)	Total Marks
1	Introduction to Biology and the Diversity of Life	5	10 (1 CRQ)	15
2	Biological Molecules and Enzymes	6	12 (1 CRQ)	18
3	Cell Structure and Function	6	12 (1 CRQ)	18
4	Cell Division and Reproduction at the Cellular Level	6	10 (1 CRQ)	16
5	Bioenergetics	7	11 (1 CRQ)	18
	Total	30	55	85

Table 2A: Overall Marks Distribution — Grade XI

Component	Certificate Entry	Marks
Theory Examination	Biology (TH)-I	85
Practical Examination	Biology (PR)-I	15
Total		100

Grade XII

Table 3: Number of Student Learning Outcomes by Cognitive Level

Topic No.	Topic	No. of Sub-topics	K	U	A	Total
6	Homeostasis and Transport	2	3	2	0	5
7	Support, Movement and Coordination	2	2	3	1	6
8	Reproduction and Development	2	2	1	1	4
9	Genetics and Molecular Biology	2	2	2	2	6
10	Evolution and Ecology	2	3	0	2	5
	Total	10	12	8	6	26
	Percentage		46%	31%	23%	100%

Table 4: Theory Exam Specification — Grade XII — Biology (TH)-II — 85 Marks

Topic No.	Topic	MCQs (Marks)	CRQs (Marks)	Total Marks
6	Homeostasis and Transport	6	11 (1 CRQ)	17
7	Support, Movement and Coordination	6	11 (1 CRQ)	17
8	Reproduction and Development	5	10 (1 CRQ)	15
9	Genetics and Molecular Biology	7	13 (1 CRQ)	20
10	Evolution and Ecology	6	10 (1 CRQ)	16
	Total	30	55	85

Table 4A: Overall Marks Distribution — Grade XII

Component	Certificate Entry	Marks
Theory Examination	Biology (TH)-II	85
Practical Examination	Biology (PR)-II	15
Total		100

- A Multiple Choice Question (MCQ) requires candidates to choose one best/correct answer from four options. Each MCQ carries ONE mark.
- A Constructed Response Question (CRQ) requires students to respond with a structured explanation, a worked calculation, a labelled diagram, or an interpretation of experimental or field data, as appropriate to the SLO being assessed.
- There will be two Theory examinations, one at the end of Grade XI (Biology (TH)-I) and one at the end of Grade XII (Biology (TH)-II). Each Theory paper carries 85 marks, is of 3 hours' duration, and consists of Paper I (MCQs) and Paper II (CRQs).
- There will be two Practical examinations, one at the end of Grade XI (Biology (PR)-I) and one at the end of Grade XII (Biology (PR)-II). Each Practical examination carries 15 marks and requires students to carry out a dissection, microscopy, or physiological investigation under supervised laboratory conditions, and to record and interpret their own results.

- The Theory and Practical marks for a given year are reported separately on the BISE Larkano Marks Certificate and are combined only in the year's subject total ($85 + 15 = 100$). A candidate must attempt both components; a Theory or Practical mark cannot substitute for the other.
- Tables 1 and 3 indicate the number and nature of SLOs in each topic and are provided as a guide to the construction of examination papers; they do not translate directly into marks. Emphasis has deliberately been placed on Application and other higher-order skills, to discourage rote memorisation of facts.
- Where a CRQ requires a numerical answer, e.g. in a genetics problem, full credit requires a correctly set-out method, not merely the final value.

Academic Integrity and Laboratory Safety

All work submitted for assessment in Biology — whether a written answer, a calculation, or a laboratory report — must represent the individual, honest effort of the student submitting it. Affiliated institutions are expected to uphold the BISE Larkano Academic Integrity Policy in full, and to make students aware of the following subject-specific guidance.

Collaboration and Use of Sources

Students are encouraged to discuss biological concepts, experimental observations and problem-solving strategies with their peers, and to consult the prescribed textbook and other reputable resources to build their understanding. However, copying a solution, a calculation, or a laboratory report from any source — including another student, a book, or an online resource — and submitting it as one's own, with or without superficial changes, is a serious violation of academic integrity.

Integrity of Experimental Data

Where practical work forms part of a student's assessment, all recorded observations and readings must be genuine. Fabricating, adjusting or selectively omitting experimental or field data to obtain an expected or more convenient result is a serious violation of academic integrity, regardless of how small the discrepancy may seem.

Laboratory Safety and the Ethical Treatment of Living Material

Affiliated institutions are responsible for ensuring that all practical work is conducted under appropriate supervision, with attention to the safe handling and disposal of biological and chemical materials, and with students made aware of relevant hazard information before an activity begins. Where practical work involves living organisms or biological specimens, institutions are expected to follow accepted standards of humane and ethical treatment, and to use alternatives to dissection where a suitable alternative achieves the same learning outcome. A student's disregard for reasonable safety or ethical instructions during practical work may be treated as a conduct matter independent of, and in addition to, their academic assessment.

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